

4. Bones can be preserved as fossils so research into evolutionary patterns and processes can include extinct vertebrates. **Image 4.1** shows how the skeleton of the modern horse has evolved from a horse-like mammal over the last 38 million years.

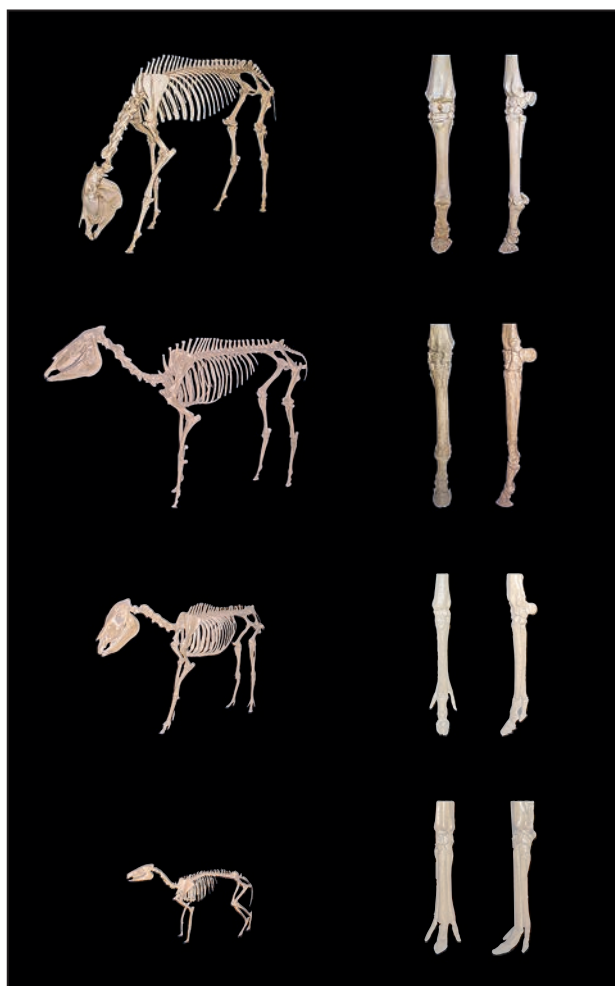
Image 4.1

Equus
Recent

Pliohippus
Late Miocene
7 MYA

Merychippus
Middle Miocene
11 MYA

Mesohippus
Late Eocene
38 MYA



MYA = Million Years ago

- (a) Explain how the information in **Image 4.1** supports Darwin's theory of evolution. [3]

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Ancient horses had many coat colours. One type of gene controlling depth of colour in horses is the dun allele (**D**). For example the bay horse in **Image 4.2** is homozygous for the **d** allele but a horse is a dun if it has at least one **D** allele (**Image 4.3**).

Image 4.2 Bay (**dd**)



Image 4.3 Dun (**DD** or **Dd**)



- (b) (i) In a sample of 900 wild horses, 891 were dun and 9 were bay. Calculate the frequencies of the three genotypes within this population, assuming that the population is in Hardy-Weinberg equilibrium. [3]

$$p + q = 1$$

$$p^2 + 2pq + q^2 = 1$$

where

p = frequency of the dominant allele (**D**)

q = frequency of the recessive allele (**d**)

Frequency of **dd** =

Frequency of **DD** =

Frequency of **Dd** =



- (ii) The population in (b)(i) consisted of randomly mating horses in semi-desert conditions.
Use all the information given to give **two** reasons why this population may not be in Hardy-Weinberg equilibrium. [2]

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Question continued overleaf



Images 4.4A and 4.4B show photographs of a mule and a hinny.

Image 4.4A – Mule



Image 4.4B – Hinny



- (c) Mules are considered more patient, hardy and long-lived than horses and are described as less obstinate and more intelligent than donkeys. They can carry heavy loads for long distances so are an ideal pack-animal.

A mule is a hybrid resulting from crossing a male donkey (*E. asinus*) with a female horse (*E. caballus*).

- (i) Explain why a breeder would need to use a donkey and a horse **each time** they wished to produce a mule. [1]

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- (ii) The opposite cross (a hinny) can also be carried out, but it is done very rarely. The mother is the much smaller donkey compared with the larger horse mother in a mule cross. Explain why the mule cross is more suitable for a load-bearing pack animal. [1]

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